



CLOUDFLARE RADAR

2024 Year in Review

Global Internet Traffic, Adoption Trends,
Connectivity Disruptions & Security Insights

Prepared for Technology Executives & Network Infrastructure Leaders

radar.cloudflare.com

17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet traffic continues upward trajectory; satellite connectivity reaches new regions

17%

Year-over-Year
Global Traffic Growth

Trend line normalized from mid-January to account for post-holiday activity

3.3

Starlink
Traffic Surge

X

SpaceX Starlink brought satellite Internet connectivity to previously underserved locations throughout 2024



Traffic growth driven by increasing reliance on the Internet for communication, commerce, entertainment & transportation

Source: Cloudflare Radar 2024 Year in Review

Google Leads Top 10 Services as Generative AI Rises

Ranking based on aggregate DNS resolver data from Cloudflare's 1.1.1.1 — generative AI saw meteoric growth in 2024

#	Service	#	Service
#1	Google	#2	Facebook
#3	Apple	#4	TikTok
#5	AWS	#6	Microsoft
#7	Instagram	#8	YouTube
#9	Amazon	#10	WhatsApp



Generative AI

Continued meteoric rise throughout 2024, reshuffling the Internet services landscape and driving unprecedented demand for GPU compute and bandwidth.

Metaverse receded into the background as enterprise and consumer focus shifted decisively toward generative AI workloads

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Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Key technology adoption metrics shaping the 2024 Internet infrastructure landscape

2% → 13%

Post-Quantum TLS 1.3

Post-quantum encrypted traffic worldwide

21%

HTTP/3 Traffic

Go

Top API Client

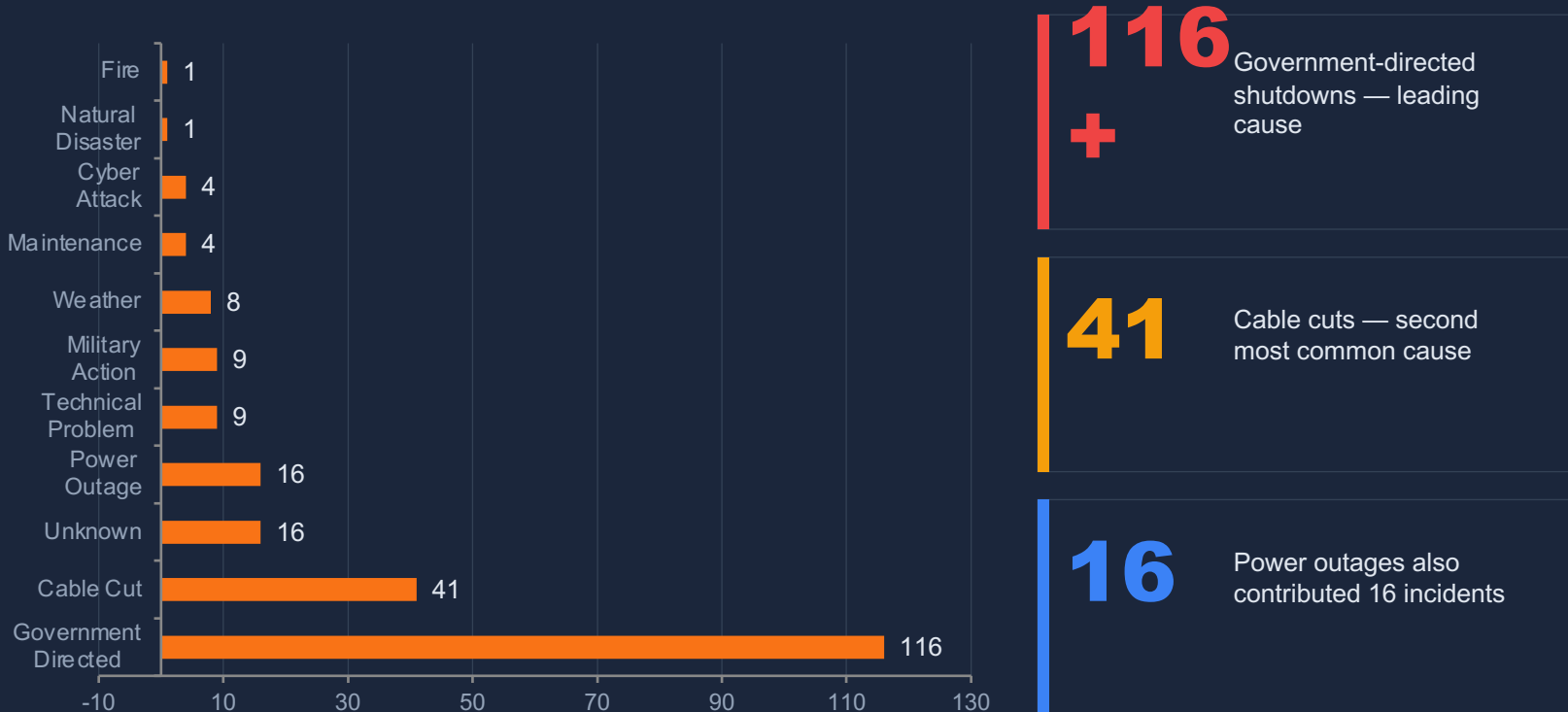


33% of traffic came from iOS devices (peak iOS ~66% in some regions)

Google search: 89% of referral traffic | Yandex 3.1% | Baidu 2.7% | Bing 2.6%

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Cloudflare tracked 225 significant Internet disruptions globally, analyzed by root cause



6.5% of Traffic Mitigated: Persistent Threats & Bot Concentration

Security landscape observed through Cloudflare's global network — threats remain pervasive

6.5%

of all traffic
was mitigated

4.3%

of emails were
malicious (43% deceptive links)

Most Attacked Sectors

1	Gambling / Games
2	Other sectors follow
3	

68% of global bot traffic originated
from the top 10 countries

⚠ Log4j

CVE-2021-44228 still actively exploited
2+ years after disclosure — persistent
nature of known vulnerabilities

+6.4%

RPKI valid routes — improved Internet
routing security

.bar TLD: largest share of malicious & spam
email origin

Five Takeaways Shaping the 2025 Internet Outlook

01

Traffic growth trajectory remains strong

17% YoY global traffic growth, with Starlink expanding 3.3x — satellite connectivity reaching underserved regions and driving new Internet usage patterns.

02

Post-quantum encryption momentum is building

TLS 1.3 post-quantum key agreement jumped from ~2% to ~13% after Chrome 124 default — momentum expected to accelerate across browsers and services.

03

Connectivity fragility from government actions

225 major disruptions observed; government-directed shutdowns (116+) remain the leading cause — cable cuts (41) and power outages (16) also significant.

04

Persistent security threats require ongoing vigilance

6.5% of all traffic mitigated; Gambling/Games most attacked; Log4j exploitation persists; 4.3% of emails malicious — attack surface continues to expand.

05

Developer ecosystem evolving rapidly

Go surpassed Node.js and Python as the top API client language (~12%); HTTP/3 at 21% adoption; generative AI reshaping the service landscape.